# Lecture 24
# State Space Techniques in CT

C.L. Wyatt

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## Introduction

Thus far we have only considered single-input, single-output (SISO) CT systems. These can be described by an N-th order LCCDE, inpulse response $h(t)$, or transfer function $H(s)$, and, if stable, via the frequency response $H(j\omega)$. In more complex systems we may be interested in multiple inputs, multiple outputs, and signals internal to the system.

Another term for signal is *state* and we actually have had multiple states in our systems, however we have removed them for analysis by reducing the system description to a SISO inpulse response or transfer function. To see this, consider a second-order LCCDE

$$a\,\frac{d^2y}{dt^2}(t) + b\,\frac{dy}{dt}(t) + c\,y(t) = x(t)$$

Recall from your differential equations course that this can also be represented by two, first-order LCCDEs. Let $u_1(t) = y(t)$ and $u_2(t) = \frac{dy}{dt} = \frac{du_1(t)}{dt}$. Then $\frac{du_2}{dt} = \frac{d^2y}{dt^2}$ and the LCCDE is

$$a\,\frac{du_2}{dt^2}(t) + b\,u_2(t) + c\,u_1(t) = x(t)$$

Solving for $\frac{du_2}{dt}$

$$\frac{du_2}{dt} = -\frac{b}{a}\,u_2(t) - \frac{c}{a}\,u_1(t) + \frac{1}{a}\,x(t)$$

Noting $\frac{du_1}{dt} = u_2$ gives the system of ODEs. Switching to dot notation for derivatives as is tradition (since we will only be dealing with first derivatives), we obtain the system of equations:

$$
\begin{aligned}
\dot{u}_1(t) &= u_2(t) \\
\dot{u}_2(t) &= -\frac{b}{a}\,u_2(t) - \frac{c}{a}\,u_1(t) + \frac{1}{a}\,x(t)
\end{aligned}
$$

Collecting $u_1$ and $u_2$ into a vector

$$
\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}
$$

we can write the system as

$$
\begin{aligned}
\dot{u} &= \begin{bmatrix} 0 & 1 \\ -\frac{c}{a} & -\frac{b}{a} \end{bmatrix} u + \begin{bmatrix} 0 \\ \frac{1}{a} \end{bmatrix} x \\
&= A\,u + B\,x
\end{aligned}
$$

where $A$ is the 2x2 matrix and $B$ is a 2x1 matrix. This form, relating first derivatives of the state to a linear combinations of the states and inputs, is called a *state equation*. Solving the state equation gives is the state/signal for both $u_1$ and $u_2$. To get the same result as solving the original LCCDE, we need the *observation equation*, in this case a single output

$$
y = [1\ 0]\,u = C\,u
$$

where $C$ is a 1x2 matrix.

To see where these internal states have been hiding consider the direct form I implementation of this system

$$
a \int \int y + b \int y + cy = x
$$

solving for $y$

$$
y = -\frac{a}{c} \int \int y - \frac{b}{c} \int y + \frac{1}{c}\,x
$$

Converting to the Laplace domain (assuming zero initial conditions

$$
Y(s) = -\frac{a}{c}\frac{1}{s^2}Y(s) - \frac{b}{c}\frac{1}{s}Y(s) + \frac{1}{c}\,X(s)
$$

As a block diagram this looks like

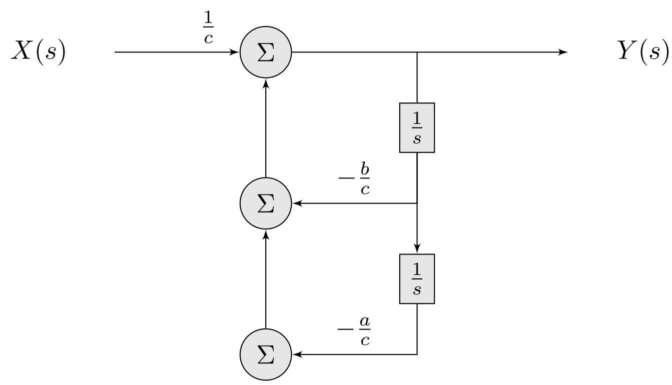


Figure 1: Block diagram for the example system.

The output of the integrator blocks correspond to the internal state variables.

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